

LAB 1: Aggregate Lab

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1. Introduction

Aggregates are important materials used in construction, especially in concrete and asphalt. Their physical properties, such as particle size distribution and specific gravity, have a major effect on the strength, durability, and overall performance of construction materials. In this experiment, sieve analysis and specific gravity tests were carried out to study these properties and assess the suitability of the aggregate samples.

2. Objectives

- To determine the particle size distribution of the aggregate samples.
- To evaluate the gradation characteristics of fine and coarse aggregates.
- To calculate the specific gravity of the aggregate.
- To assess whether the aggregates are suitable for construction applications.

3. Theory

Sieve Analysis

Sieve analysis is a testing method used to determine the distribution of particle sizes within an aggregate sample. The sample is passed through a series of standard sieves arranged from the largest opening size to the smallest. The mass retained on each sieve is measured and used to calculate the percentage passing. These results help determine whether the aggregate is well graded or poorly graded.

The uniformity coefficient and coefficient of curvature are calculated using:

$$C_u = D_{60} / D_{10}$$

$$C_c = (D_{30})^2 / (D_{10} \times D_{60})$$

Specific Gravity

Specific gravity is the ratio of the density of an aggregate to the density of water. It is used to evaluate the relative density and quality of the material.

$$SG = W_3 / [W_3 - (W_1 - W_2)]$$

4. Diagram of Apparatus

5. Experimental Procedure

Sieve Analysis Procedure

The required maximum aggregate size was first used to select the appropriate sieves according to the testing requirements. Before starting the test, all sieves were cleaned properly to avoid contamination from particles left from previous tests.

The sieves were arranged in descending order, with the largest sieve at the top and the smallest sieve above the collection pan. A representative aggregate sample was then weighed accurately using a digital balance.

The measured sample was placed into the top sieve, and the complete sieve stack was secured in the mechanical sieve shaker. The shaker was operated continuously for approximately 10 minutes to separate the particles according to size.

After shaking was completed, the sieves were removed carefully, and the mass of aggregate retained on each sieve was recorded. Any particles stuck to the sieves were gently brushed off to ensure accurate weighing. The cumulative mass retained was then calculated, followed by the mass passing through each sieve. Finally, the percentage passing for each sieve size was determined using the total sample mass.

Specific Gravity Procedure

The coarse aggregate sample was washed thoroughly to remove dust and fine particles that could affect the accuracy of the test. The cleaned aggregate was placed in a wire basket and fully immersed in water for 24 hours to allow it to become saturated.

The basket containing the aggregate was weighed while submerged in water, and this value was recorded as W1. The weight of the empty basket in water was then measured and recorded as W2.

After this, the aggregate was removed and dried with absorbent cloths until it reached a saturated surface dry condition. The surface-dry aggregate was weighed in air, and this value was recorded as W3. These measured values were then substituted into the specific gravity formula to calculate the aggregate's specific gravity.

6. Results

Specific Gravity Data

W1 = 2329 g W2 = 1082 g W3 = 2003 g

Specific Gravity = 2.65

Fine Aggregate

Cu = 10.67 Cc = 1.50 Classification: Well graded

Coarse Aggregate

Cu = 1.88 Cc = 0.83 Classification: Poorly graded

7. Graphical Presentation

Figure 3: Grading Curve for Fine Aggregate

D60 = 1.6 mm D30 = 0.6 mm D10 = 0.15 mm

Figure 4: Grading Curve for Coarse Aggregate

D60 = 15 mm D30 = 10 mm D10 = 8 mm

Appendix 1: Tabulated Results

Table 1.1: Sieve Analysis Results for Fine Aggregates, 3000 g

Sieve Size	Empty Sieve Mass	Empty + Soil	Retained	Cumulative Retain~	Passing	Passing %
2.36	1180.3	1863.5	683.2	683.2	2316.8	77.23
2.0	1186.7	1399.0	212.3	895.5	2104.5	70.15
1.0	1002.7	1832.3	829.6	1725.1	1274.9	42.50
0.63	1013.3	1377.4	364.1	2089.2	910.8	30.36
0.315	855.5	1190.4	334.9	2424.1	575.9	19.20
0.16	748.8	970.9	222.1	2646.2	353.8	11.79
Pan	714.3	1065.1	350.8	2997.0	0	0

Table 1.2: Sieve Analysis Results for Coarse Aggregates, 5000 g

Sieve Size	Empty Sieve Mass	Empty + Soil	Retained	Cumulative Retain~	Passing	Passing %
31.5	-	-	0	0	5000	100
25	-	-	0	0	5000	100
16	1387	3120	1733	1733	3267	65.34
5.6	1086	4233	3147	4880	120	2.40
Pan	714.3	835.8	121.5	5001.5	0	0

Appendix 2: Calculations

Example Sieve Calculation

Mass retained = 1863.5 - 1180.3 = 683.2 g

Cumulative retained = 683.2 + 212.3 = 895.5 g

Mass passing = 3000 - 895.5 = 2104.5 g

Percentage passing = (2104.5 / 3000) x 100 = 70.15%

Specific Gravity Calculation

W1 - W2 = 2329 - 1082 = 1247 g

W3 - (W1 - W2) = 2003 - 1247 = 756 g

SG = 2003 / 756 = 2.65

Fine Aggregate Calculations

Uniformity Coefficient:

Cu = D60 / D10

$$C_u = 1.6 / 0.15 = 10.67$$

Coefficient of Curvature:

$$C_c = (D_{30})^2 / (D_{10} \times D_{60})$$

$$C_c = (0.6)^2 / (0.15 \times 1.6)$$

$$C_c = 0.36 / 0.24 = 1.50$$

Coarse Aggregate Calculations

Uniformity Coefficient:

$$C_u = D_{60} / D_{10}$$

$$C_u = 15 / 8 = 1.875, \text{ approximately } 1.88$$

Coefficient of Curvature:

$$C_c = (D_{30})^2 / (D_{10} \times D_{60})$$

$$C_c = (10)^2 / (8 \times 15)$$

$$C_c = 100 / 120 = 0.833, \text{ approximately } 0.83$$

8. Analysis and Discussion

The calculated specific gravity of the aggregate is 2.65, which falls within the typical range for construction aggregates. This indicates that the material has a suitable density and can be considered appropriate for construction use.

The sieve analysis results show that the fine aggregate has a good range of particle sizes. Its uniformity coefficient is 10.67, and its coefficient of curvature is 1.50, which falls within the acceptable range of 1 to 3. This means the fine aggregate is well graded. A well-graded aggregate helps improve compaction, reduce void spaces, and increase the strength and durability of construction materials.

In comparison, the coarse aggregate has a lower uniformity coefficient of 1.88 and a coefficient of curvature of 0.83. Since the coefficient of curvature is less than 1, the coarse aggregate is considered poorly graded. Poorly graded aggregates usually contain a limited range of particle sizes, which can lead to higher void content and reduced stability. Therefore, the coarse aggregate may need to be blended with other sizes to achieve better gradation.

Overall, the fine aggregate is more suitable for construction use based on its grading properties, while the coarse aggregate may require modification before being used in structural applications.

9. Conclusions

The aggregate samples were tested to determine their particle size distribution and specific gravity. The specific gravity value of 2.65 indicates that the aggregate has a suitable density for construction purposes.

The fine aggregate was found to be well graded, making it suitable for use in construction materials where good compaction and strength are required. The coarse aggregate was classified as poorly graded because it had a limited particle size distribution. As a result, it may need to be combined with other aggregate sizes to improve its grading.

In conclusion, the experiment showed that specific gravity and gradation are important properties

for evaluating the performance and suitability of aggregates in engineering applications.

10. References

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